Missing Angles on a Straight Line

Year 6 · Mathematics · Measurement | C2 · Learn + Notice

Name: _____ Date: _____

LEARNING INTENTION

I can calculate unknown angles using a 180° total.

KEY IDEA

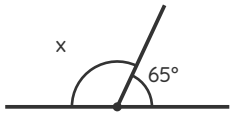
Adjacent angles that fill a straight angle add to 180°. Add all known parts, then subtract from 180°.

VOCABULARY

straight angle · adjacent · supplementary · unknown

LEARN IT: A Worked Model

A straight angle is split into 65° and x. Write 65° + x = 180°. Therefore x = 115°. Check: 65° + 115° = 180°.



Use the marked opening. Diagrams for calculation tasks are not drawn to scale.

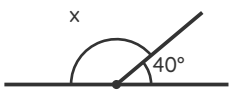
SUCCESS CRITERIA

- [ ] I can calculate unknown angles using a 180° total.
- [ ] I can explain the benchmark, tool or angle rule I used.
- [ ] I can check my answer and correct a mistake.

NOTICE IT: Guided Practice

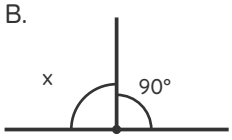
Find x. Write the rule or total you used.

A.



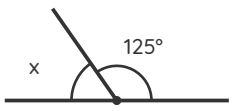
Answer: _____

B.



Answer: _____

C.



Answer: _____

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Year 6 · Mathematics · Measurement | C2 · Practise + Apply + Reason

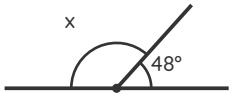
PRACTISE IT: Independent Questions

Find x. Write the rule or total you used.

A.

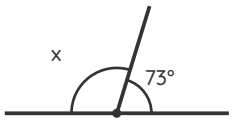
Answer: _____

B.



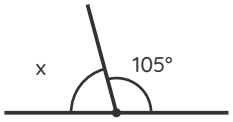
Answer: _____

C.



Answer: _____

D.

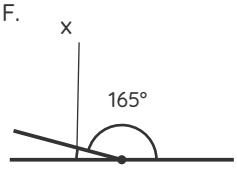


Answer: _____

E.

Answer: _____

F.



Answer: _____

APPLY IT

A straight angle is split into 35°, 60° and x. Find x and write a check equation.

EXPLAIN HOW YOU KNOW: Misconception Check

Ruby says any two angles drawn next to one another add to 180°. What extra condition is needed?

CHALLENGE: Reasoning Task

A straight angle has two equal parts and a third part of 40°. Find each equal part.

MY SELF-REVIEW

Circle one: I understand it / I need more practice / I need help

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ANSWER KEY: Guided + Independent Practice

Notice: A 140°; B 90°; C 55°.

Practice: A 155°; B 132°; C 107°; D 75°; E 48°; F 15°.

Apply: 85°; 35° + 60° + 85° = 180°.

Reason: the parts must fill a straight angle, with outer arms forming a straight line.

Challenge: each equal part is 70°; (180° - 40°) / 2 = 70°.

COMMON MISCONCEPTION

Adjacent angles are supplementary only when together they form a straight angle.

TEACHER PROMPT

Ask: "Which opening are you using? What benchmark or angle relationship applies? How can you check your answer?"

NEXT-STEP GUIDANCE

Move to unknown angles around a point.

PRINTING AND ASSESSMENT

Print A4. Diagrams are not to scale for calculation purposes. Use stated values and angle relationships; check the explanation as well as the result.

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